

End-to-End Customer Analytics and Churn Prediction System

Project Phases and Work Plan

Phase 1: Data Understanding

We will explore the dataset, check number of rows and columns, understand data types, identify target column, and detect missing values.

Phase 2: Data Cleaning

We will handle missing values, remove duplicate records, correct data types and prepare the dataset for analysis.

Phase 3: Exploratory Data Analysis (EDA)

We will analyze patterns such as customer age vs churn, country vs churn, purchases vs churn and login frequency vs churn.

Phase 4: Data Visualization

We will create bar charts, boxplots, distribution plots and correlation heatmaps to understand relationships.

Phase 5: Feature Engineering

We will create new features like engagement level, customer value segments and risk categories.

Phase 6: Machine Learning

We will build churn prediction models such as Logistic Regression, Decision Tree and Random Forest.

Phase 7: Model Evaluation

We will measure model performance using accuracy, confusion matrix, precision, recall and F1 score.

Phase 8: SQL Integration

We will store cleaned data in MySQL and perform business queries such as top customers and churn statistics.

Phase 9: Dashboard Creation

We will create Power BI dashboards with KPIs like total customers, churn rate and average lifetime value.

Phase 10: Flask API Deployment

We will create a Flask API to allow real-time churn prediction using customer data input.

Final Outcome

At the end of this project we will have a clean dataset, analytical insights, a trained churn prediction model, business dashboard, SQL queries and a real-time prediction API. This demonstrates a complete industry-level data science workflow.